



Cyclistic Bike-Share Case Study

Understanding our customers to drive Membership growth.

Google Data Analytics Capstone Project
by Tomas Iglesias Campaña
2026

The Company



Cyclistic is a bike-share program that operates in the Chicago area with a fleet of 5824 geotracked bicycles and 692 stations.

The company offers a variety of pricing plans: single-ride passes, full-day passes and annual memberships.

Objectives



The objective of this analysis is to understand the customer base and find opportunities to convert casual riders into annual members. We want to find out:

- How do annual members and casual riders differ?
- Why would casual riders buy Cyclistic annual memberships?
- How can Cyclistic use digital media to influence casual riders to become members?

The Data

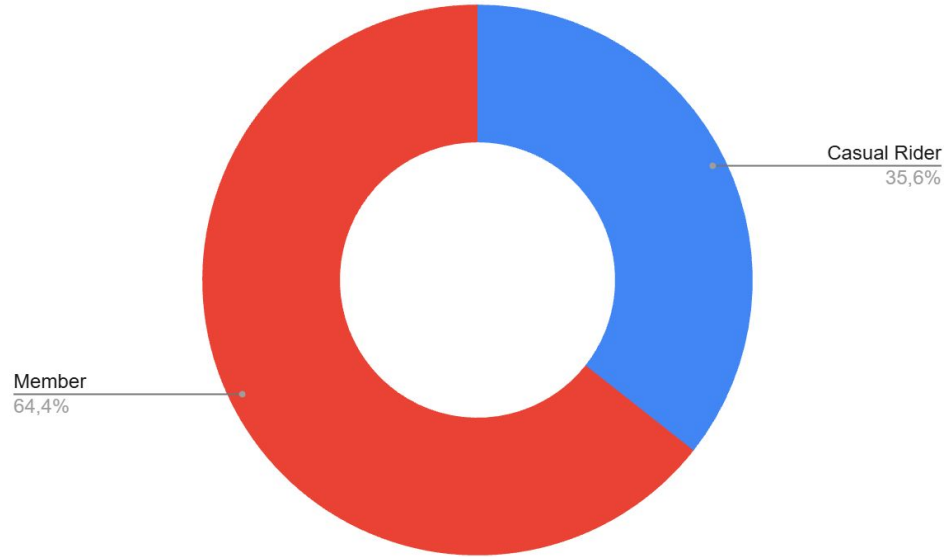


For this analysis we are using the last 12 months of trip data, from August 2025 to July 2026. It's provided directly from the company, passing all ROCCC criteria.

The data was joined together and all incomplete or inconsistent values were removed. Rows with null values were removed, as well as negative or less than 60 seconds ride duration values.

This left us with a total of 3,9 million valid ride entries.

Analysis // Riders

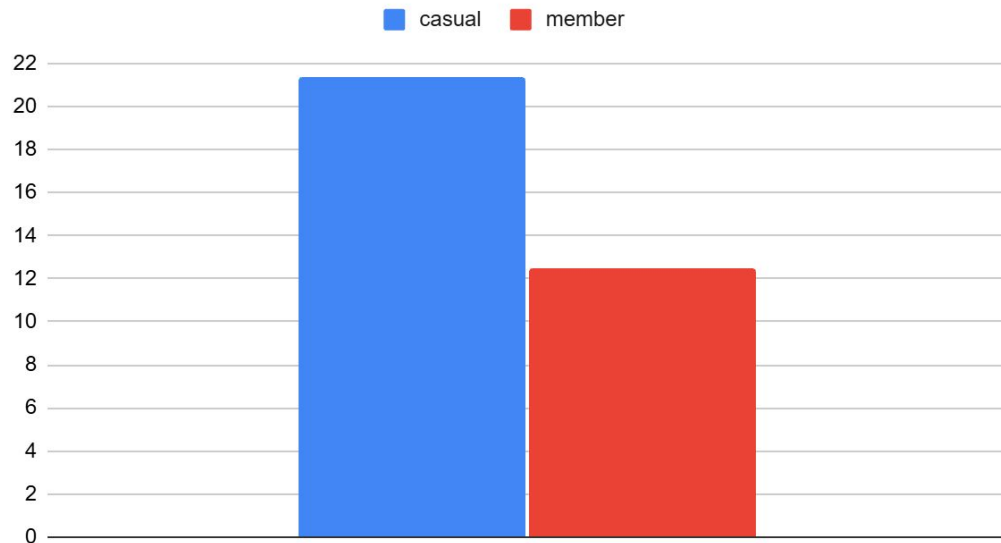


First we want to know who our customers are.

Around two thirds of the rides are made by Cyclistic members, and the other third come from casual riders. Our focus is to find out how these casual riders use the service differently.

Analysis // Riders

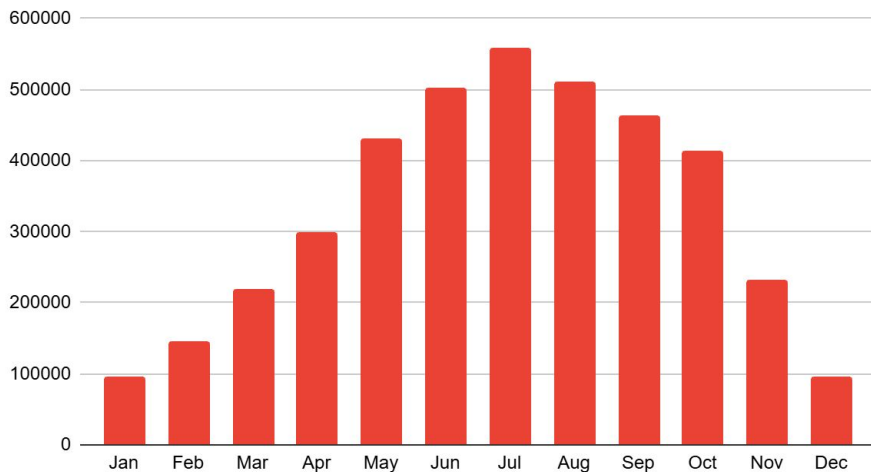
Ride Duration in Minutes by Type



We can observe that casual riders tend to use the bikes for significantly longer periods than members.

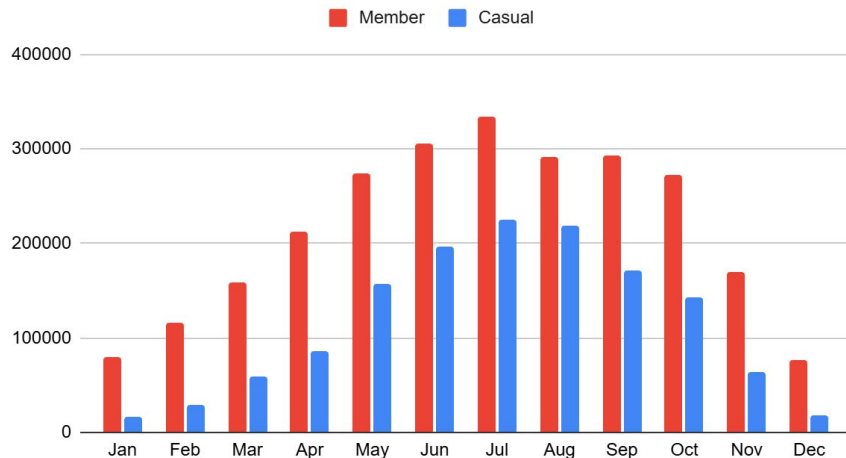
Analysis // Year

Riders by Month



The usage of the service varies significantly by season, with the most usage during summer and very low usage in winter.

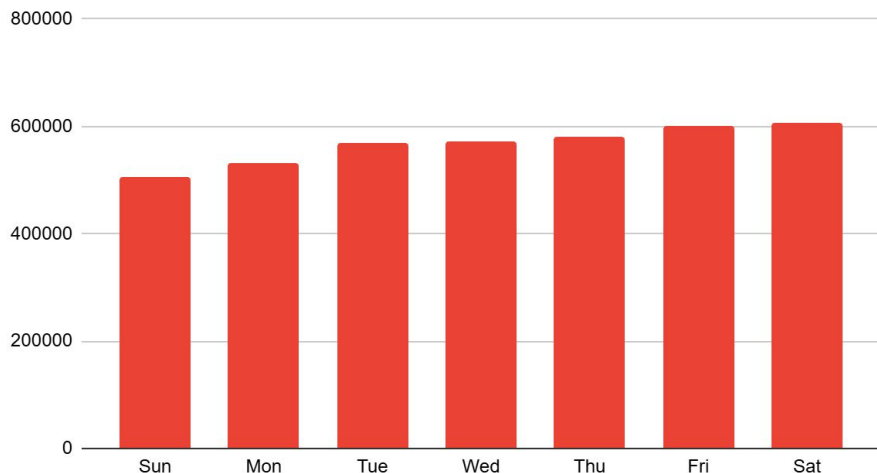
Type of Rider by Month



When divided by type of rider, we see that casual riders are mostly present during the summer, while members are more consistent through the year.

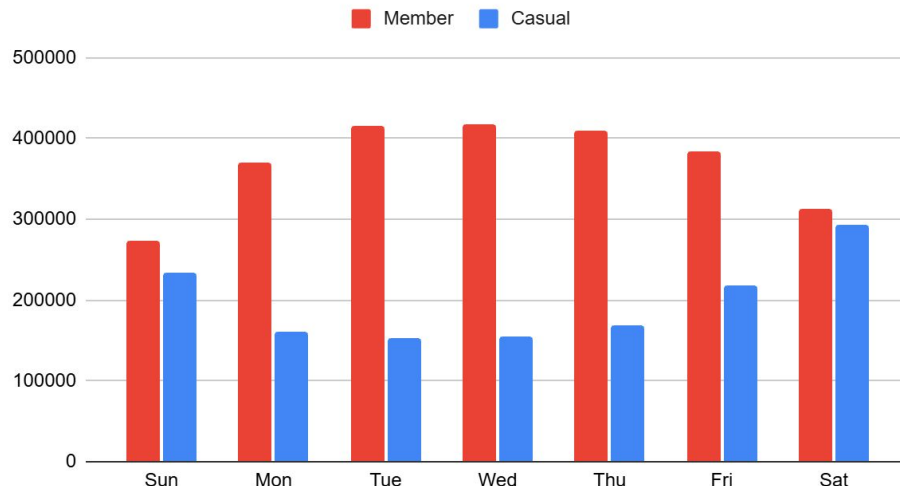
Analysis // Week

Riders by Day of the Week



Usage between days of the week increases towards the weekend, but doesn't have a very important difference.

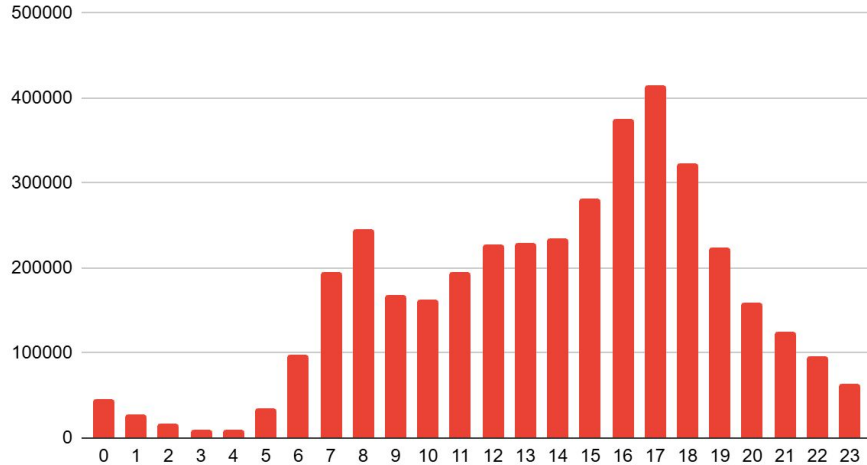
Riders by Day of the Week



Again, we see a difference in rider types. Members mostly use the service during the week, while casual riders use it for weekends.

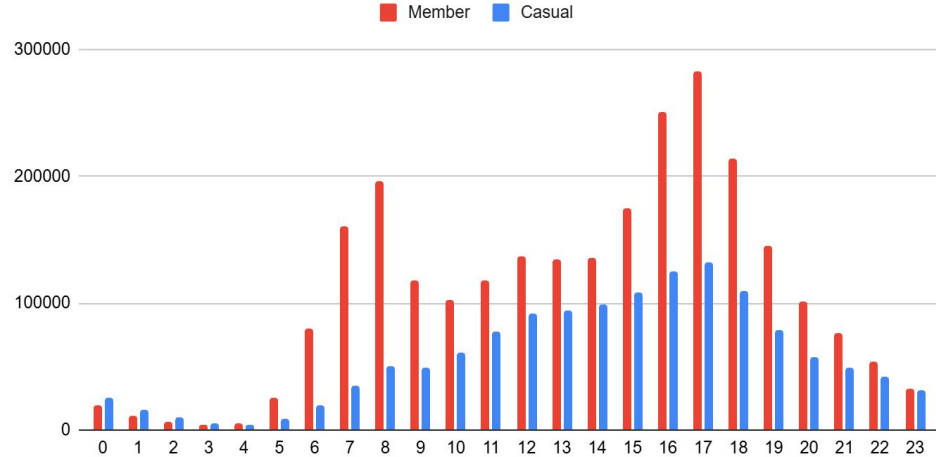
Analysis // Day

Rides by Hour of the Day



The total rides by hour of the day show a small peak at morning rush hour, as well as a much larger peak at the late afternoon.

Type of Rider by Hour of the Day



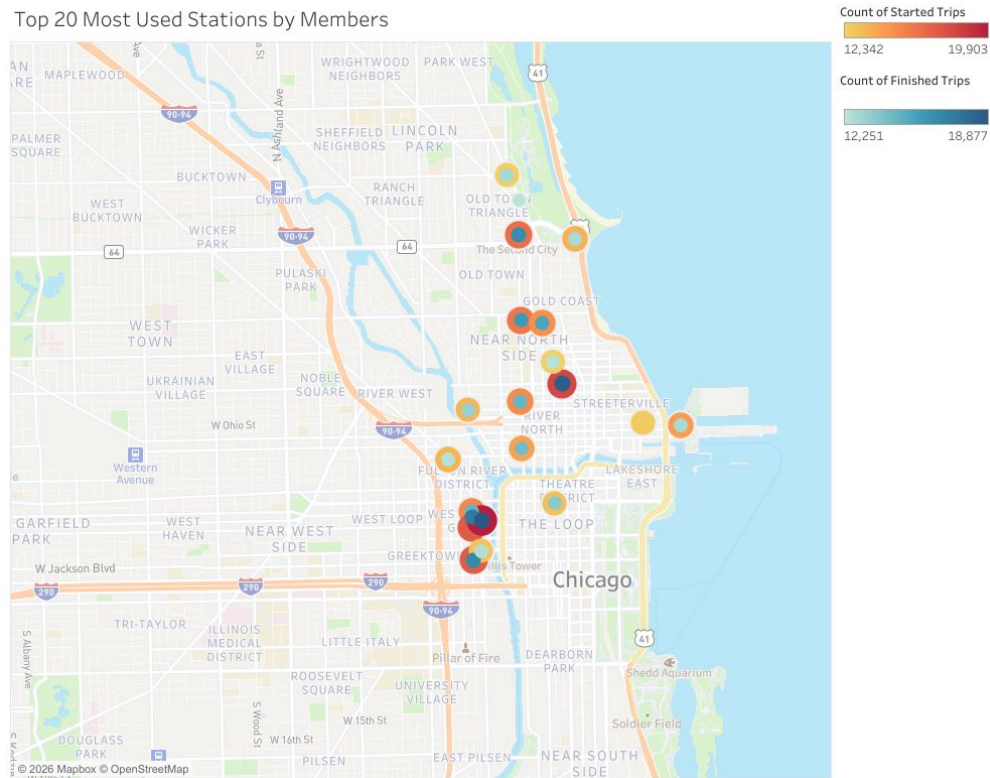
Members peak during morning and afternoon rush hours, while casual users use it mostly in the afternoon.

Analysis // Stations

When visualizing the top 20 most used stations by members, we see a clear pattern of density around the main financial centers of the city.

There is a big concentration around the West Loop, the main financial center of the city, and many more points going along the important office districts of the city.

Top 20 Most Used Stations by Members

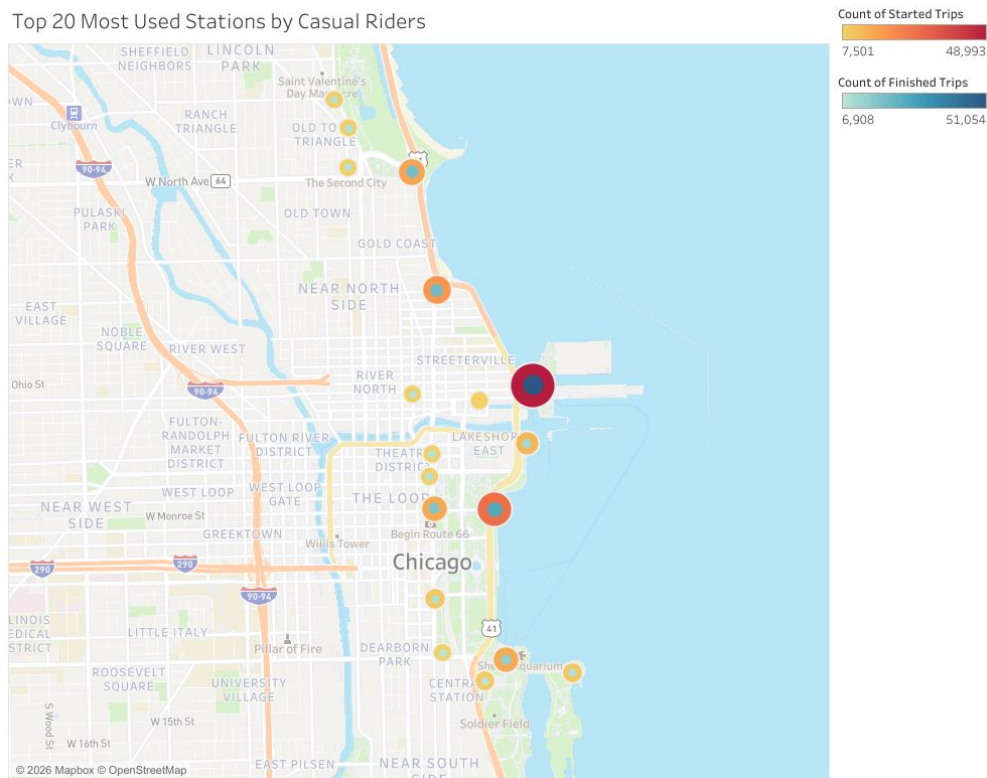


Analysis // Stations

On the other hand, when visualizing the most used stations by casual riders, we see a completely different pattern.

The stations concentrate along the lake shore and major parks, zones that suggest that casual riders are more likely to use the service for leisure.

Top 20 Most Used Stations by Casual Riders



Conclusion



After reviewing all of the information, we found these key insights:

- Casual riders represent 36% of the rides.
- Their ride duration is significantly longer than members'.
- They use the service mostly during the summer period and on weekends.
- They concentrate around lake shore and public park areas.

We can conclude that the data strongly suggests that while members mostly use Cyclistic as a means of transportation, particularly during weekdays and commuting hours, casual riders show usage patterns more consistent with leisurely rides.

Understanding this key insight into the riders is essential to finding the best strategy to increase the number of members using the service.

Recommendations

Considering all of this information, we have 3 proposals to drive up the number of memberships:

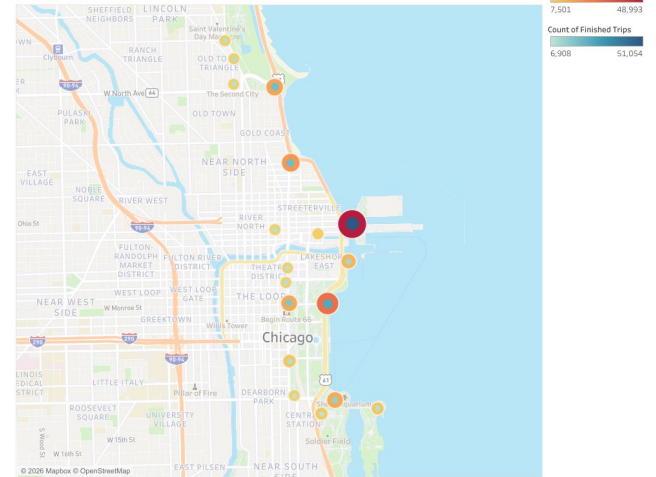
- **Targeted Marketing Campaign:**

The data collected provides a strong base to launch a marketing campaign targeted to the casual riders. We can roll out a strong campaign on the April - May months, with additional focus on weekends, just before the high season to catch as many new customers as possible.

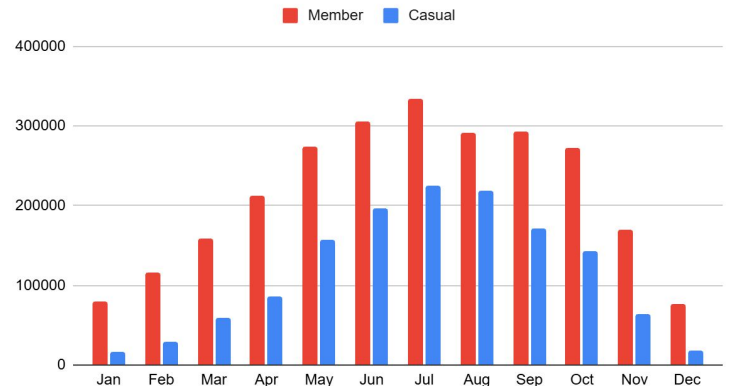
We can also focus the campaign on public parks and lake shore areas, where the data shows the casual riders are more concentrated.

The campaign could be focused on the extent of the stations network, specially on the areas we mentioned are most busy for these riders.

Top 20 Most Used Stations by Casual Riders



Type of Rider by Month



Recommendations

- **Collection of additional data:**

Compiling an additional dataset on customer accounts with individual information on user age range, number of trips and approximate location could provide additional insights that could help design a more precise marketing campaign.

It could also provide an answer to exactly how many individual customers are members and how many are casual riders.

Recommendations



- New membership options:

Considering that casual riders mostly use the service on summer months and weekends, creating a new summer exclusive or weekend exclusive membership at a more accessible price could earn us a significant amount of new members.

This would be the most radical option, but with additional data on customers and pricing we could determine if it's financially viable.



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